

P11.1: $P=(1,0,0) \in \mathbb{R}^3$; $x+y-z=0$

(i) Geom.

$$\vec{n} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\vec{r}(t) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1+t \\ t \\ -t \end{pmatrix}$$

→ Schnittpkt:

$$1+t + t - (-t) = 0 \Rightarrow 1+3t=0$$

$$t_0 = -\frac{1}{3} \Rightarrow \vec{r}_0 = \begin{pmatrix} 2/3 \\ -1/3 \\ 1/3 \end{pmatrix}$$

$$d = \sqrt{\left(\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2}$$

$$d = \frac{1}{\sqrt{3}}$$

(ii) $x+y=z$ | N.B.

Abstands fkt: $f(x,y,z) = (x-1)^2 + y^2 + z^2$

$$\tilde{f}(x,y) = (x-1)^2 + y^2 + (x+y)^2$$

$$\nabla \tilde{f}(x,y) = \begin{pmatrix} 2(x-1) + 2(x+y) \\ 2y + 2(x+y) \end{pmatrix} \stackrel{!}{=} \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$4x + 2y - 2 = 0 \Rightarrow 3x + x + 2y - 2 = 0$$

$$\frac{4y}{2} + \frac{2x}{2} = \frac{0}{2} \Rightarrow 2y + x = 0 \Rightarrow 3x - 2 = 0$$

$$y = -1/3$$

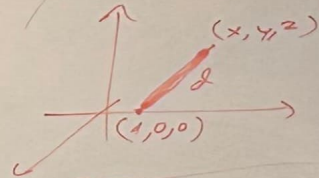
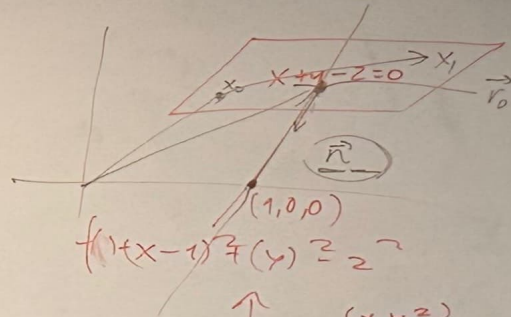
$$\Rightarrow z = x+y = 1/3$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2/3 \\ -1/3 \\ 1/3 \end{pmatrix}$$

$$H_f(x,y) = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$$

$$\det H_f(x,y) = 12 > 0$$

$$\text{Tr } H_f(x,y) = 8 > 0$$



$$\partial_1 d = \sqrt{(x-1)^2 + (y)^2 + (z-0)^2}$$

$$\begin{pmatrix} -4 & -4 \\ -4 & -4 \end{pmatrix} = 16$$

(iii) $f(x, y, z) = (x-1)^2 + y^2 + z^2$

N.B.: $z = x + y$

$\exists \lambda$ (Voraussetzung):

$$\nabla f(x, y, z) = \lambda \cdot \nabla g(x, y, z)$$

$$\begin{pmatrix} 2(x-1) \\ 2y \\ 2z \end{pmatrix} = \lambda \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\begin{aligned} 2(x-1) &= \lambda \Rightarrow x = \frac{\lambda}{2} + 1 \\ 2y &= \lambda \Rightarrow y = \frac{\lambda}{2} \\ 2z &= -\lambda \Rightarrow z = -\frac{\lambda}{2} \end{aligned}$$

$f - \lambda$

$$\frac{\lambda}{2} + 1 + \frac{\lambda}{2} + \frac{\lambda}{2} = \frac{3\lambda}{2} = -1 \Rightarrow \lambda = -\frac{2}{3}$$

$$x = -\frac{1}{3} + 1 = \frac{2}{3}$$

$$y = -\frac{1}{3}$$

$$z = \frac{1}{3}$$

$$\varphi^{-1}(x, y, 0) = (x, y, h(x, y)) \in G_h$$

$\varphi^{-1} \in \mathcal{C}^k(U \times \{0\})$
 $\varphi \in \mathcal{C}^k$

$\Rightarrow \varphi$ ein Diffeomorphismus! , $\dim G_h = 2 \Rightarrow G_h$ 2-dim $\mathcal{C}^k$ -UMF von $\mathbb{R}^3$

11.2: $U \subseteq \mathbb{R}^2$ offen, $h \in \mathcal{C}^k(U)$

(a) zz: $G_h = \{(x, y, h(x, y)) \in \mathbb{R}^3 \mid (x, y) \in U\}$ eine 2-dim $\mathcal{C}^k$ -UMF von $\mathbb{R}^3$ ist.

Bew: Wir definieren:

$$\varphi: V = U \times \mathbb{R} \rightarrow \mathbb{R}^3, \varphi(x, y, z) = (x, y, z - h(x, y)) \quad \mathbb{R}^m \times \{0\}^{n-m}$$

Es gilt:

$$\varphi(G_h) = \{(x, y, 0) \in \mathbb{R}^3 \mid (x, y) \in U\} = U \times \{0\}^{(3-2)}$$

$$(x, y, 0) \in U \times \{0\}$$

$$\varphi^{-1}(x, y, 0) = (x, y, 0)$$

$$(b) \psi: G_h \rightarrow U, \psi(x, y, h(x, y)) = (x, y)$$

$$\psi^{-1}: U \rightarrow G_h, \psi^{-1}(x, y) = (x, y, h(x, y))$$

$$(c) \text{ Sei } f: V \rightarrow \mathbb{R}, f(x, y, z) = \underline{z - h(x, y)}$$

$$f(G_h) = f(x, y, h(x, y)) = \underline{0} \quad \frac{(x, y, h(x, y))}{(x, y, h(x, y))}$$

$$G_h = f^{-1}(\{0\})$$

$$J_f(x, y, z) = \begin{pmatrix} -\partial_x h(x, y) & -\partial_y h(x, y) & 1 \end{pmatrix}$$

$$\rightarrow \underline{\text{rang } J_f(x, y, z) = 1 \quad \forall (x, y) \in U.}$$

Beh. aus dem Satz 2 (R.W.) $\square$

$$P11.3: M := \{(x, y, z) \in \mathbb{R}^3 \mid z > 0 \wedge z^2 = x^2 + y^2\}$$

(a) z.z.: M eine 2-dim C^∞ -UMF des $\mathbb{R}^3$.

Ben.: $U := \{(x, y, z) \in \mathbb{R}^3 \mid \underline{z} > 0\}$ „obere Halbraum“

$$\text{und: } f: U \rightarrow \mathbb{R}, f(x, y, z) = x^2 + y^2 - z^2$$

$$M \cap U = \{(x, y, z) \in U \mid x^2 + y^2 = z^2\} = \{(x, y, z) \in U \mid f(x, y, z) = 0\}$$

$$J_f(x, y, z) = \begin{pmatrix} 2x & 2y & -2z \end{pmatrix} \quad z \neq 0$$

$$\boxed{\text{rang}(J_f(x, y, z)) = 1}$$

U eine ~~offen~~ Umg. $\forall x \in M$

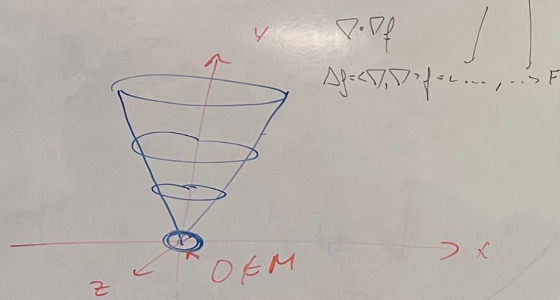
S.R.W. $\Rightarrow$ M ist C^∞ -UMF von $\mathbb{R}^3$ (2-dim).

Alternativ C^∞ -Fkt. $h: \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}, h(x, y) = \sqrt{x^2 + y^2}$

$$(x, y, z) \rightarrow (\rho, \varphi, z)$$

$$\underline{x^2 + y^2 = \rho^2 = z^2}$$

$$\nabla f = \left(\frac{\partial f}{\partial r}, \frac{1}{r} \frac{\partial f}{\partial \theta}, \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \right)$$



(c) $\overline{M} = \{(x, y, z) \in \mathbb{R}^3 \mid z \geq 0, z^2 = y^2 + x^2\}$

$$y_f(x, y, z) = (2x \quad 2y \quad -2z)$$

$$\text{rang } \mathcal{Y}_f(x, y, z) \neq 1 \quad \forall (x, y) \in \bar{M}.$$

$$f(x, t) \quad F(x, y) = \int_0^{y_1} f(x, t) dt \quad T_F^2 \text{ im } (0, 0)$$

①. $F(0,0) = \int_0^{5(0)-0} f(x,t) dt = \underline{0}$

$$(2): \partial_x F(x, y) = \partial_x \int_0^{y_1} f(x, t) dt = \int_0^{y_1} \partial_x f(x, t) dt$$

$$\partial_x F(0, \rho) = 0$$

④: $\partial_x \partial_x F(a, 0) = 0$

(5): $\partial_1 \partial_2 F(x, y) = \partial_2 \partial_1 F(x, y) = \frac{\partial_2}{0} \int_0^{5y} \partial_1 f(x, t) dt = \partial_1 f(x, 5y) \cdot 5$
 $\partial_2 \partial_1 F(0, 0) = 5 \partial_1 f(0, 0)$

$$F(y) = \int^{u(y)} f(t) dt$$

$$\partial_y F(y) = f(u(y)) \cdot u'(y)$$

$$\begin{aligned} 7) &= \partial_2 5 f(x, 5y) \\ &= 5 \left(\partial_x f(x, 5y) \frac{\partial x}{\partial y} \right) = 0 \\ &\quad + \partial_2 f(x, 5y) \cdot 5 \\ &= 5 \partial_2 f(x, 5y) \end{aligned}$$

$$\frac{\partial}{\partial x} f(5xy, y) \cdot 5x + \frac{\partial}{\partial y} f(5xy, y) \cdot 1$$